MANIPAL UNIVERSITY JAIPUR

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School of Computer Science and Engineering
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PATH PULSE

Emphasizes awareness and intelligence in daily movement

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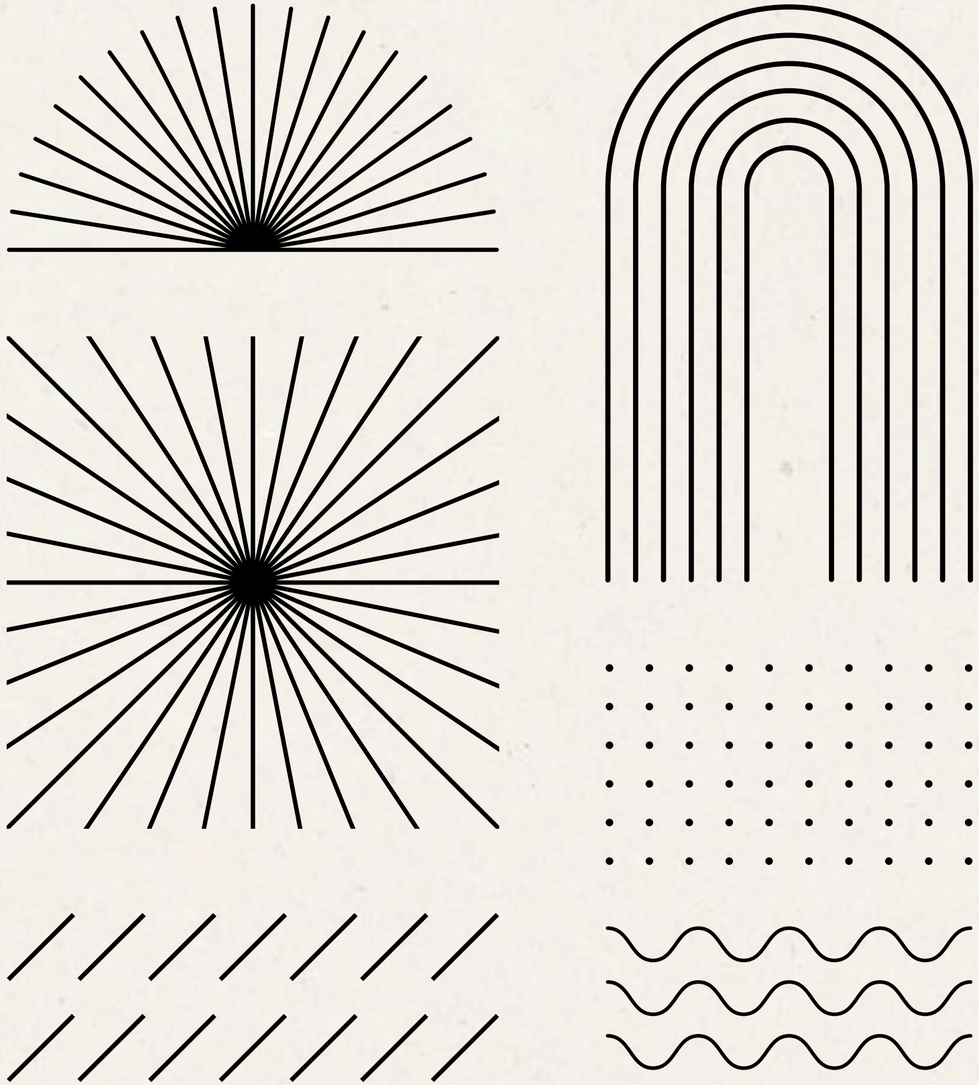
OUTLINE

- INTRODUCTION
- LITERATURE REVIEW
- PROBLEM STATEMENT
- PROPOSED SOLUTION
- OBJECTIVE
- REFERENCES



INTRODUCTION

01 Women’s safety during daily travel has become a major concern, especially when commuting alone or at late hours. Similar risks can also affect children, elders, and other individuals..

02 Many people follow fixed daily routes, and any sudden change or delay can signal potential danger. Tracking such movements can help identify unsafe situations early.

03 This app aims to enhance safety by detecting unusual travel patterns and ensuring quick alerts to trusted contacts when something seems wrong.

LITERATURE REVIEW

- Noonlight (formerly SafeTrek) :Emergency alerts, hold-button /silent SOS features.
- Life360:Real-time location sharing, geofencing, crash ,detection, emergency contacts. Could notice when user leaves expected zone
- Existing safety apps rely on manual trigger;few use behavioral learning for proactive alerts.
- This project fills the gap by combining routine learning with real-time alerting, tailored to individual movement patterns.
- Time-series models like LSTM and clustering algorithms(K-Means, DBSCAN) are effective for identifying deviations in sequential data.

Paper Name	Abstract	Technology Used	Result	Conclusion
Routine Pattern Discovery and Anomaly Detection in Individual Travel Behavior (Sun et al., 2023)	Explores spatiotemporal travel behavior and discovers routine movement patterns. Uses probabilistic modeling to detect anomalies in daily travel routes.	2D-LDA (Two-Dimensional Latent Dirichlet Allocation), unsupervised learning on trajectory/LPR datasets.	Successfully identifies unusual trips by comparing them with learned daily patterns.	Probabilistic topic models work well in capturing sparse human travel behavior and detecting deviations.
Mobile Trajectory Anomaly Detection: Taxonomy, Methodology, Challenges, and Directions (Kong et al., 2024)	A complete survey of anomaly detection in mobile trajectories. Defines anomaly types and compares common algorithms.	Deep learning, clustering, density-based models, sequential modeling, mobility data processing.	Provides a unified taxonomy and shows strengths/weaknesses of existing models.	Future systems must focus on privacy, real-time detection, sparse-data handling, and interpretability.
Threat Eye – Safety Alert System for Real-Time Threat Detection	Develops an Android app that sends rapid SOS messages and shares live location when the user is in danger.	Android Studio, GPS, SMS/Internet communication, mobile UI.	Successfully sends emergency alerts and shares real-time location with trusted contacts.	Smartphone-based alert systems improve personal safety by enabling quick emergency communication.
Feasibility of Anomalous Event Detection Using Mobile Spatial Statistics (Mas & Koshimura, 2024)	Examines whether aggregated mobile location data can detect population anomalies related to events/disasters.	Mobile Spatial Statistics (MSS), spatial-temporal anomaly detection.	Found that unusual spikes/drops in population movement align with known events.	MSS is useful for large-scale anomaly detection but requires calibration for biases and local patterns.
Anomaly Detection & Mobile Network Security Enhancement Using K-Means on CDRs (Aziz & Bestak, 2024)	Uses clustering on call detail records to detect suspicious or unusual mobile network behavior.	K-means clustering, CDR analytics, large-scale data processing.	Achieved high anomaly detection accuracy using telecom data patterns.	CDR-based clustering is effective for detecting network threats and security anomalies.

PROBLEM STATEMENT

In today’s world, personal safety has become a growing concern, especially for individuals who travel alone or at odd hours. Existing safety apps rely on manual alerts, which may not be possible during emergencies. There is a need for an intelligent system that can automatically learn a person’s daily travel routine, detect unusual route deviations or timing anomalies in real time, and alert trusted contacts without requiring user intervention.

OBJECTIVE

- To develop a smart safety application that ensures quick and reliable emergency response for users.
- To enhance personal safety, especially for women and vulnerable individuals, through AI and real-time monitoring.
- To detect unusual or risky situations automatically using machine learning and location tracking.
- To enable hands-free emergency alerts using voice activation and custom code words.
- To build community awareness and cooperation through a connected safety network.

METHODOLOGY

SYSTEM ARCHITECTURE & TECHNOLOGIES

METHODOLOGY – SYSTEM ARCHITECTURE

• The system is developed as a web-based safety application using Flask.

• It follows a client–server architecture.

Frontend:

- HTML, CSS, Bootstrap
- JavaScript for location handling
- Responsive dashboard interface

Backend:

- Python Flask framework
- Handles location processing and anomaly detection

Database:

- SQLite database (routes.db)
- Stores latitude, longitude, and timestamp

APIs Used:

- Browser Geolocation API
- Google Maps for live location display

Real-Time Location Collection & Storage

METHODOLOGY – LOCATION TRACKING

- The system collects the user's real-time location using the Geolocation API.

Steps:

- User opens the Dashboard
- Browser requests location permission
- Latitude and longitude are captured automatically
- Location is displayed on:
- Dashboard coordinates
- Google Map view
-

Data Storage:

- Each location is sent to Flask backend
- Backend stores data in SQLite with timestamp
- This creates a continuous travel history

Routine Learning & Anomaly Detection

Proposed Methodology – Intelligent Monitoring

- The system first learns the user's normal travel behavior.
- Routine Learning
- Initial location records are collected
- System calculates the average latitude and longitude
- This average represents the routine location area
- Anomaly Detection
- Each new location is compared with routine location
- Distance is calculated using coordinate difference
- If distance exceeds predefined threshold → SOS Status
- Otherwise → SAFE Status

This enables automatic abnormal movement detection without manual input.

Alert Mechanism & User Interface

Proposed Methodology – Response & Visualization

Alert System:

- SAFE → Green status displayed
- SOS → Red alert with warning message
- Automatic detection based on location deviation

User Interface Features

- Modern dashboard with gradient design
- Live Google Map display
- Real-time coordinates
- Safety status indicator
- Location history page
- Navigation between Home, Dashboard, and History

Outcome

A complete real-time personal safety monitoring system with automated risk detection and history tracking.

GeoLife Dataset – Detailed Description

- The system uses the GeoLife GPS Trajectory Dataset developed by Microsoft Research
- It is a real-world dataset containing human mobility patterns collected over 5 years (2007–2012)

Dataset Features:

- Contains over 17,000+ GPS trajectories
- Collected from multiple users
- Captures real movement across urban environments

Data Attributes:

- Latitude (geographical coordinate)
- Longitude (geographical coordinate)
- Timestamp (date and time)
- Optional: altitude, speed

Data Format:

- Stored in .plt files
- Each entry represents a location point:
  Latitude, Longitude, Altitude, Date, Time

Why this Dataset?

- Provides realistic GPS movement data
- Widely used in trajectory mining and anomaly detection research
- Enables safe simulation without privacy concerns

Dataset Processing & Integration

Data Pipeline:
GeoLife (.plt files) → Python Processing → SQLite Database → Flask Application

Processing Steps:
1. File Handling
   - .plt files are read using Python
   - First 6 lines (metadata) are ignored
2. Data Extraction
   - Extracted fields:
     - Latitude
     - Longitude
     - Timestamp
3. Data Storage
   - Stored in SQLite database (routes.db)
   - Table structure:

(latitude, longitude, time)

Integration with System:
- Dataset simulates historical user movement
- Used to build travel history records
- Supports training phase of anomaly detection

Importance in Project:
- Eliminates dependency on real-time user data
- Enables testing of system behavior under real conditions
- Provides continuous location data for learning patterns

Model Used – Anomaly Detection Approach

- The system uses a rule-based anomaly detection model
- Designed to identify deviations from normal travel behavior

Working of the Model:

1. Routine Learning Phase

- Initial location data is collected
- System computes:

Average Latitude (avg_lat)

Average Longitude (avg_lon)

- Represents normal travel region

2. Detection Phase

- For each new incoming location:

Distance = $\sqrt{((lat - avg_lat)^2 + (lon - avg_lon)^2)}$

3. Decision Rule

- If distance ≤ threshold → SAFE
- If distance > threshold → SOS ALERT

Implementation Details:

- Implemented using Python (Flask backend)
- Real-time location processed via API
- Threshold-based classification
- Lightweight and fast execution

Advantages:

- Simple and efficient
- Real-time processing
- No heavy training required
- Easily scalable

Limitations & Future Scope:

- Does not consider time-based patterns
- Fixed threshold may vary across users
- Future improvements:
  - K-Means clustering
  - LSTM-based sequence learning

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THANK YOU